

Trainings ▾

General Information

Some engines are only built with a camshaft bushing installed in the first camshaft bore on the same side of the engine that the camshaft is removed and installed (front gear train engines, at the front of the engine and rear gear trains, at the rear of the engine). Some engines also have a second camshaft bushing at the opposite end of the engine from which the camshaft was removed and installed.

From production, there are no camshaft bushings installed in the inner block camshaft journal bores. If the inner block camshaft journal bores are damaged, the journal bores can be machined for camshaft bushings. See the Clean and Inspect for Reuse step of this procedure for specifications.

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Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

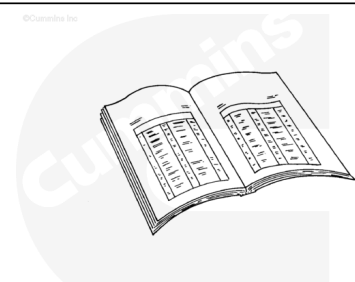
⚠ WARNING ⚠

To avoid the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged

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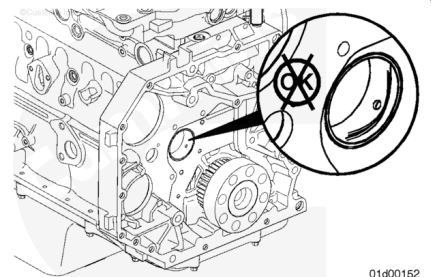
contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Remove the camshaft. Refer to Procedure 001-008 (Camshaft) in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html)
- Drain the lubricating oil. Refer to Procedure 007-037 (Lubricating Oil System) in Section 7.
(/qs3/pubsys2/xml/en/procedures/40/40-007-037.html)
- Remove the oil pan and oil pan gasket. Refer to Procedure 007-025 (Lubricating Oil Pan) in Section 7.
(/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html)

Initial Check

Inspect the camshaft bushing at the end of the engine in which the camshaft was removed for scoring, scuffing or excessive wear.

If damage to the camshaft bushing is found, remove and replace the camshaft bushing.



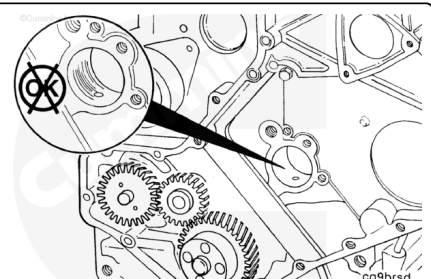
From below the engine, inspect if a camshaft bushing is installed at the opposite end of the engine in which the camshaft was removed.

If a camshaft bushing is present, inspect the camshaft bushing for scoring, scuffing or excessive wear.

If damage to the camshaft bushing is found, remove and replace the camshaft bushing.

To replace the camshaft bushing, you must first:

For Rear Gear Train Engines:



Remove the front gear cover. Refer to Procedure 001-031 (Gear Cover, Front) in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html)

For Front Gear Train Engines:

Remove the flywheel housing. Refer to Procedure 016-006 (Engine Component Torque Values) in Section 16.
(/qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html)

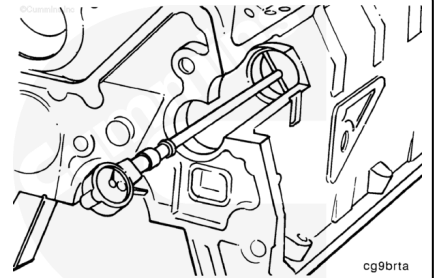
From below the engine, inspect the inner camshaft bores for scoring, scuffing or excessive wear.

Note : During a previous repair, camshaft bushings may have been installed.

For camshaft bore specifications see the Inspect for Reuse step of this procedure.

If damage to the inner camshaft bores is found, machine the camshaft bores oversize to install standard camshaft bushings. See the Inspect for Reuse step for specifications.

If the damage to the bore(s) is beyond machining, replace the block. Refer to Procedure 001-026 (Cylinder Block) in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-026.html)



Remove

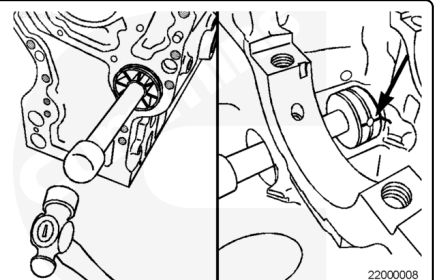
Remove the camshaft bushings, use the camshaft bushing replacer kit, Part Number 3165045. The tool will drive the camshaft bushing to the inside of the block. The camshaft bushing can be retrieved from the bottom of the engine.

The following kits are also available for camshaft bushing installation and removal:

Hydraulic Actuator Kit, Part Number 3823621

Camshaft Bushing Kit, Part Number 3162249

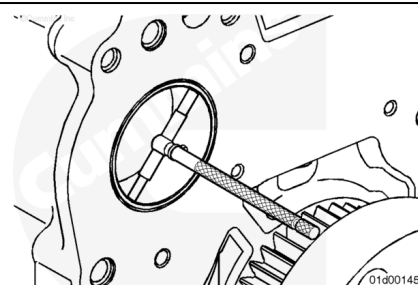
Note : Once removed, camshaft bushings can not be reused.



Inspect for Reuse

Rear Gear Train

Measure the camshaft bores **without** the camshaft bushing installed.



	mm		in	
Camshaft Bore (Camshaft Bushing Previously Installed)		59.248	MAX	2.3326
Camshaft Bore (Camshaft Bushing not Previously Installed)		54.164	MAX	2.1324

If a camshaft bushing was previously installed and the camshaft bore is out of specification, the block **must** be replaced. Refer to Procedure 001-026 (Cylinder Block) in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-026.html>) No oversize cam bushings are available.

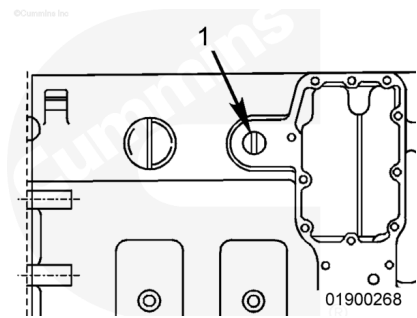
If a camshaft bore in which a camshaft bushing was **not** previously installed is damaged or out of specification, the camshaft bore can be machined oversize to install a standard camshaft bushing.

Machine the damaged bore to a maximum of 59.248 mm (2.3326 in) to accept a standard bushing.

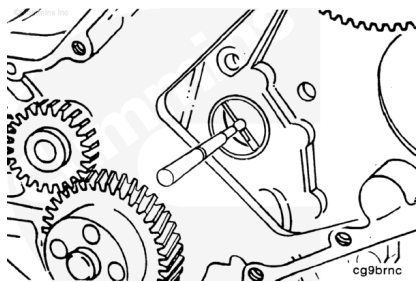
Front Gear Train

With the introduction of the STORM block, the camshaft bushing bore increased by 2 mm [0.079 in] to accept a larger outer diameter camshaft bushing. The camshaft bushing inner diameter did **not** change.

Note : A STORM block will have a threaded port next to the oil cooler cavity in the cylinder block (1). A non-STORM block will **not** have this port.



Measure the camshaft bores **without** the camshaft bushing installed.



	mm		in	
Camshaft Bore (Camshaft Bushing Previously Installed)		59.248	MAX	2.3326
Camshaft Bore (Camshaft Bushing not Previously Installed)		54.164	MAX	2.1324

	mm		in	
Camshaft Bore (Camshaft Bushing Previously Installed)		57.248	MAX	2.2539

Camshaft Bore (Camshaft Bushing not Previously Installed)		54.164	MAX	2.1324
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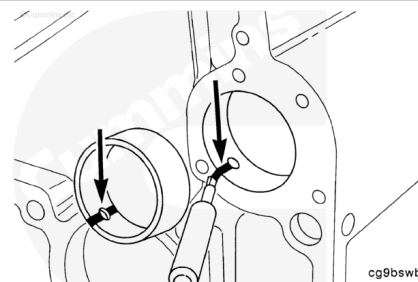
If a camshaft bushing was previously installed and the camshaft bore is out of specification, the block **must** be replaced. Refer to Procedure 001-026 (Cylinder Block) in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-026.html>) No oversize camshaft bushings are available.

If a camshaft bore in which a camshaft bushing was **not** previously installed is damaged or out of specification, the camshaft bore can be machined oversize to install a standard camshaft bushing.

Machine the damaged bore to a maximum of 59.248 mm (2.3326 in) to accept a standard bushing.

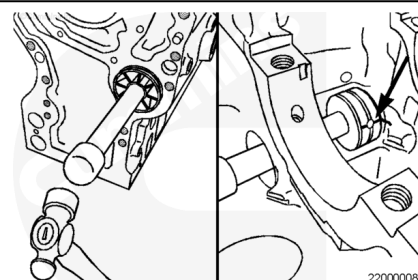
Install

Mark the camshaft bushing and cylinder block to aid in the alignment of the lubricating oil hole in the cylinder block with the large lubricating oil hole in the bushing.



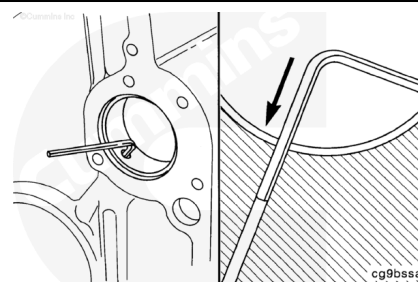
Slide the camshaft bushing on the replacer tool. Align the marks on the camshaft bushing and the cylinder block.

Drive the camshaft bushing to the correct installed depth. The correct installed depth is when the camshaft bushing end is flush with the machined face of the block and the oil hole aligns with the cylinder block oil hole.



Be sure the large lubricating oil hole is aligned.

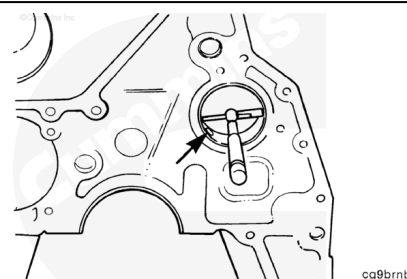
A 3.2 mm [0.128 in] diameter rod **must** be able to pass through the lubricating oil hole.



Measure

Measure the installed camshaft bushing.

mm		in
54.083	MIN	2.1293
54.147	MAX	2.1318

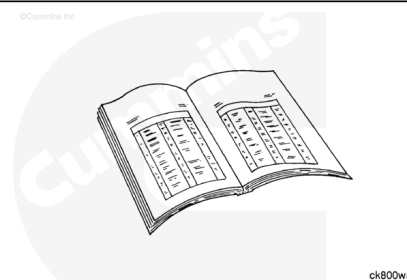


If the camshaft bore with the camshaft bushing installed is out of specification, remove the camshaft bushing and inspect the camshaft bore.

Finishing Steps

⚠ WARNING ⚠

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- Install the camshaft. Refer to Procedure 001-008 (Camshaft) in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html)
- Install the oil pan gasket and oil pan. Procedure 007-025 (Lubricating Oil Pan) in Section 7.
(/qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 (Lubricating Oil System) in Section 7.
(/qs3/pubsys2/xml/en/procedures/40/40-007-037.html)
- Connect the batteries. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine and check for leaks.